

Lab-04: Classification of Skin Lesions

Using ResNet-34 (from Scratch) & ResNet-152 (Pretrained)

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Dataset: ISIC 2020 Challenge Dataset

Introduction

This lab implements binary melanoma classification from dermoscopy images using two ResNet architectures. ResNet-34 was built entirely from scratch using basic PyTorch primitives (`Conv2d`, `BatchNorm2d`, `ReLU`, `MaxPool2d`, `AdaptiveAvgPool2d`, `Linear`). ResNet-152 used pretrained ImageNet weights loaded into our custom architecture. Both models output a single binary logit (benign vs. melanoma).

Dataset & Custom Dataset Class

Dataset Overview

The ISIC 2020 dataset contains 33,126 labelled dermoscopy images with severe class imbalance:

- Benign (class 0): 32,542 images — 98.24%
- Melanoma (class 1): 584 images — 1.76%

Due to this imbalance, raw accuracy is a misleading metric. **AUC-ROC** was used as the primary evaluation metric throughout.

Data Split

The ISIC official test set has no ground truth labels, so a stratified split of the labelled data was used (`random_state=42`):

Split	Size	Purpose
actual_train	~23,850	Weight updates each epoch
monitor_val	~2,650	Val AUC per epoch, checkpoint saving
test_held_out	~6,626	Locked — opened once for final evaluation

Custom Dataset & Imbalance Handling

A custom `SkinLesionDataset` class (extending `torch.utils.data.Dataset`) loads JPEG images and applies transforms. To handle class imbalance:

- `WeightedRandomSampler` — oversamples minority class in each training batch
- `pos_weight` ≈ 55 in `BCEWithLogitsLoss` — penalises missed melanoma cases 55× more
- Optimal threshold via Youden's J statistic: $\arg \max(\text{TPR} - \text{FPR})$

Model Architectures

Model	Block	Config	Parameters	Weights
ResNet-34 (Scratch)	BasicBlock	[3,4,6,3]	21,285,185	Kaiming Normal
ResNet-152 (Pretrained)	Bottleneck	[3,8,36,3]	58,156,993	ImageNet (V2)

ResNet-34 uses BasicBlock units with two 3×3 convolutions and a skip connection. Parameter count was verified to be 21,285,185, consistent with a binary-head ResNet-34 (original 1000-class head has $512 \times 999 = 509,488$ additional parameters).

ResNet-152 uses Bottleneck blocks ($1 \times 1 \rightarrow 3 \times 3 \rightarrow 1 \times 1$, expansion=4). Pretrained weights were loaded key-by-key with 468/468 keys matched. The FC head was replaced with a single binary output neuron. A **differential learning rate** was applied: backbone at 10^{-5} ($10 \times$ lower) to preserve ImageNet features, head at 10^{-4} .

Training Setup & Experiments

Common Configuration

Loss: BCEWithLogitsLoss with `pos_weight`≈55 · Batch size: 32 · Input: 224×224 RGB · AMP + gradient clipping (`max_norm`=1.0) · Augmentation: HorizontalFlip, VerticalFlip, Rotation(20°), ColorJitter.

Experiment Configurations

Exp	Model	LR	Optimizer	Scheduler	Best Val AUC
1	ResNet-34 (Scratch)	10^{-3}	Adam	None	0.8202
2	ResNet-34 (Scratch)	10^{-4}	Adam	StepLR (step=4, $\gamma=0.1$)	0.8134
3	ResNet-34 (Scratch)	10^{-4}	AdamW (wd= 10^{-4})	CosineAnnealing	0.8144
4	ResNet-152 (Pretrained)	10^{-5} / 10^{-4}	AdamW (wd= 10^{-4})	CosineAnnealing	0.8602

Experiment Observations

Experiment 1 (Best Val AUC: 0.8202) — LR= 10^{-3} was too high for training from scratch. Loss oscillated in early epochs due to unstable gradient updates. Without a scheduler, the LR never adapted, limiting convergence.

Experiment 2 (Best Val AUC: 0.8134) — Reducing LR to 10^{-4} gave smoother training curves. StepLR improved convergence but caused performance stalls at each discrete $10 \times$ drop. Slightly lower than Exp 1 due to abrupt drops near the final epochs.

Experiment 3 (Best Val AUC: 0.8144) — AdamW fixes the weight decay decoupling issue in standard Adam. CosineAnnealing provides smooth LR decay, allowing broad exploration early and fine-grained refinement later. Best scratch-trained result.

Experiment 4 (Best Val AUC: 0.8602) — Pretrained ResNet-152 achieved a high AUC from epoch 1, confirming that ImageNet features (textures, edges) transfer well to dermoscopy images. Differential LR was essential — without it, pretrained features would be overwritten by gradient signals from the randomly-initialised head.

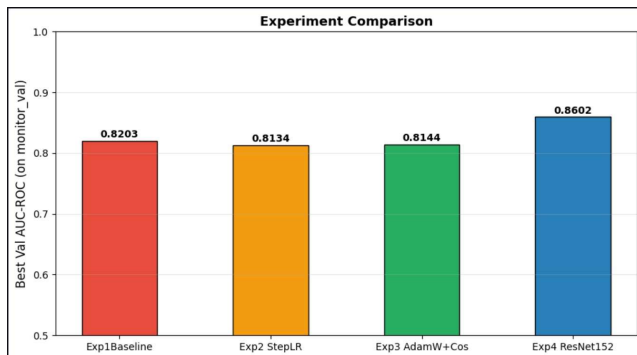


Figure 1: Best validation AUC-ROC comparison across all four experiments.

Final Evaluation on test_held_out

All four models were evaluated on the locked `test_held_out` set (6,626 images, 117 melanoma cases, 1.76%). Optimal threshold per model was determined using Youden's J statistic.

Val AUC vs. Test AUC — All Experiments

Exp	Model	Val AUC	Test AUC
1	ResNet-34 Baseline	0.8202	0.812
2	ResNet-34 StepLR	0.8134	0.813
3	ResNet-34 AdamW+Cosine	0.8144	0.804
4	ResNet-152 Pretrained	0.8602	0.888

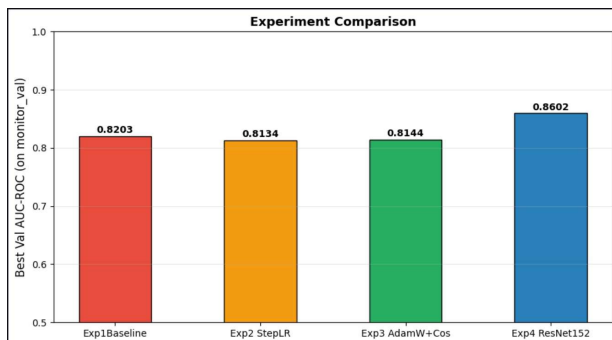


Figure 2: Best validation AUC-ROC comparison across all four experiments.

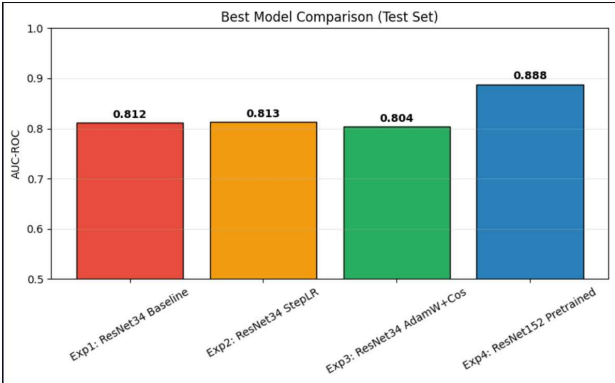


Figure 3: Test set AUC-ROC comparison across all four experiments.

Full Metrics — All Experiments on test_held_out

Model	AUC-ROC	Accuracy	Bal. Acc	Sensitivity	Specificity	F1 (w)	Precision	Kappa
Exp1: ResNet-34 Baseline	0.8116	0.5998	0.7627	0.9316	0.5938	0.7328	0.0396	0.0436
Exp2: ResNet-34 StepLR	0.8131	0.6847	0.7556	0.8291	0.6821	0.7968	0.0448	0.0533
Exp3: ResNet-34 AdamW+Cos	0.8039	0.6200	0.7562	0.8974	0.6150	0.7487	0.0402	0.0447
Exp4: ResNet-152 Pretrained	0.8881	0.7867	0.8327	0.8803	0.7851	0.8653	0.0686	0.0977

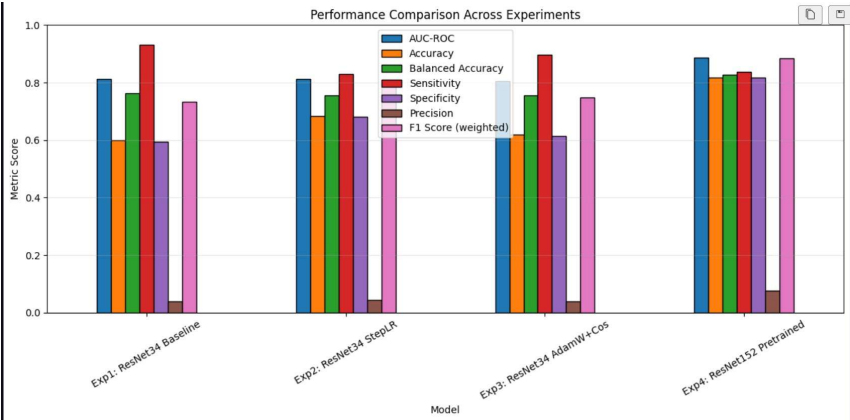


Figure 4: Per-metric performance comparison across all four experiments on test_held_out.

Detailed Metrics — Experiment 4 (ResNet-152 Pretrained)

Threshold = 0.1536 (Youden's J).

Metric	Value	Interpretation
AUC-ROC	0.8881	88.8% probability of ranking melanoma above benign
Accuracy	0.7867	Misleading due to class imbalance
Balanced Accuracy	0.8327	Average of sensitivity & specificity
Sensitivity	0.8803	103/117 melanoma cases correctly detected
Specificity	0.7851	5110/6509 benign cases correctly identified
F1 (weighted)	0.8653	Strong overall classification quality
Precision	0.0686	Low due to 117 positives in 6,626 images
Cohen's Kappa	0.0977	Penalised by FP from aggressive threshold

Confusion Matrices (all experiments):

Model	TP	TN	FP	FN
Exp1: ResNet-34 Baseline	109	3865	2644	8
Exp2: ResNet-34 StepLR	97	4440	2069	20
Exp3: ResNet-34 AdamW+Cos	105	4003	2506	12
Exp4: ResNet-152 Pretrained	103	5110	1399	14

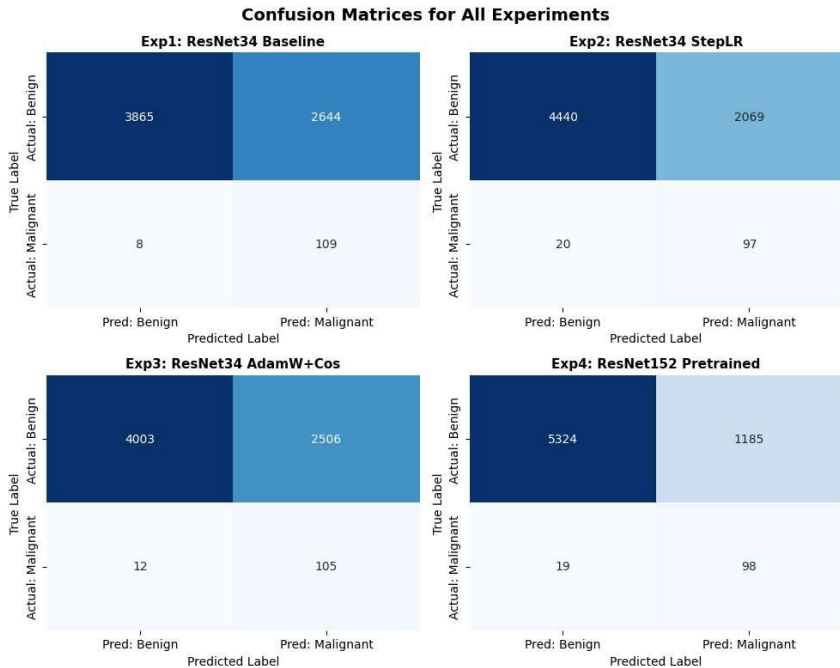


Figure 5: Confusion matrices for all four experiments on test_held_out.

Metric Analysis

What went well:

- **AUC-ROC = 0.8881** — Excellent. The model has an 88.81% probability of ranking a melanoma image higher than a benign one. This is the most reliable metric here.
- **Sensitivity = 0.8803** — The model caught 103 out of 117 real melanoma cases (only 14 missed). Clinically, this is the most important number — missing a cancer is far worse than a false alarm.
- **Balanced Accuracy = 0.8327** — Good performance across both classes, not just the majority benign class.

What looks concerning (and why it is expected):

- **Precision = 0.0686** — Out of 1,502 images flagged as melanoma, only 103 were actually melanoma. This is a direct consequence of the class imbalance: there are only 117 true positives in 6,626 test images (1.76%). The model is being aggressive by design (`pos_weight=55`), which inflates false positives.
- **Cohen's Kappa = 0.0977** — Near-zero agreement, heavily penalised by the 1,399 false positives. This metric does not suit highly imbalanced tasks where the decision threshold is intentionally shifted.
- **FP = 1,399** — 1,399 benign cases wrongly flagged as melanoma. The threshold of 0.1536 is very low (driven by `pos_weight=55`), causing the model to flag aggressively.

Why this is acceptable: The low precision and kappa are a known trade-off with imbalanced medical datasets. In melanoma screening, *missing a real cancer (FN) is clinically far worse than a false alarm (FP)* — false positives simply proceed to further dermatologist review. AUC-ROC = 0.8881 and Sensitivity = 0.8803 confirm that the model is genuinely learning to discriminate melanoma, not just predicting by chance.

Summary & Learnings

- **Transfer learning** was the single biggest performance lever — pretrained ResNet-152 outperformed all scratch-trained models from epoch 1.
- **Differential LR** is essential for fine-tuning: backbone at $10\times$ lower LR preserves pretrained features.
- **AdamW + CosineAnnealing** (Exp 3) is the best-practice combination for training CNNs from scratch.
- **AUC-ROC** is the correct primary metric for imbalanced datasets — raw accuracy is deceptive at 1.76% class frequency.
- For melanoma screening, **high sensitivity at the cost of precision** is the correct clinical trade-off: missing a cancer is worse than a false alarm.